

Mind’s Eye of LLMs: Visualization-of-Thought Elicits Spatial Reasoning in Large Language Models

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Abstract

Large language models (LLMs) have exhibited impressive performance in language comprehension and various reasoning tasks. However, their abilities in spatial reasoning, a crucial aspect of human cognition, remain relatively unexplored. Human possess a remarkable ability to create mental images of unseen objects and actions through a process known as **the Mind’s Eye**, enabling the imagination of the unseen world. Inspired by this cognitive capacity, we propose Visualization-of-Thought (**VoT**) prompting. VoT aims to elicit spatial reasoning of LLMs by visualizing their reasoning traces, thereby guiding subsequent reasoning steps. We employed VoT for multi-hop spatial reasoning tasks, including natural language navigation, visual navigation, and visual tiling in 2D grid worlds. Experimental results demonstrated that VoT significantly enhances the spatial reasoning abilities of LLMs. Notably, VoT outperformed existing multimodal large language models (MLLMs) in these tasks. While VoT works surprisingly well on LLMs, the ability to generate *mental images* to facilitate spatial reasoning resembles the mind’s eye process, suggesting its potential viability in MLLMs. Please find the dataset and codes in our [project page](#).

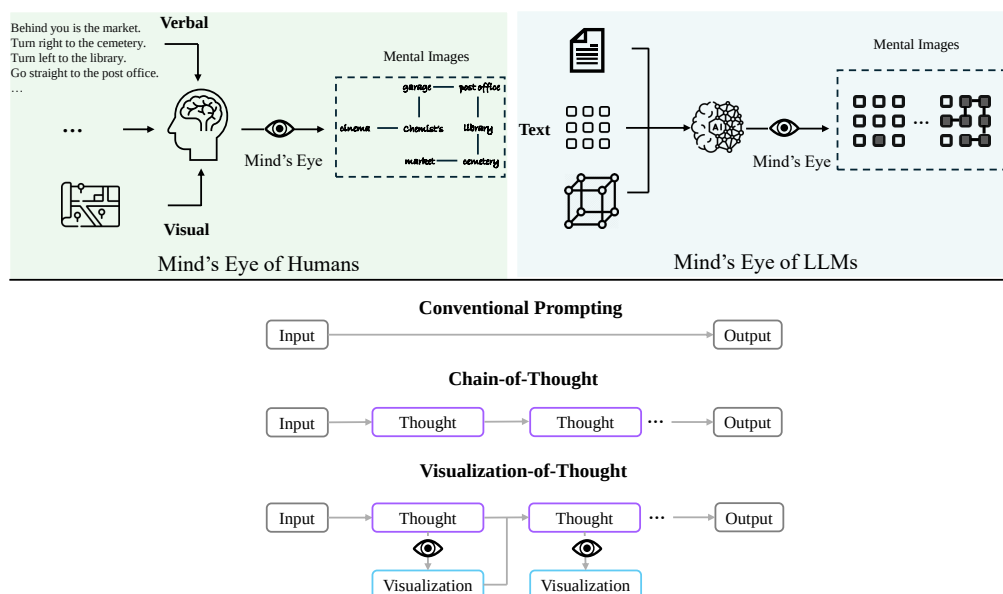


Figure 1: Humans can enhance their spatial awareness and inform decisions by creating mental images during the spatial reasoning process. Similarly, large language models (LLMs) can create internal *mental images*. We propose the VoT prompting to elicit the "mind’s eye" of LLMs for spatial reasoning by visualizing their thoughts at each intermediate step.

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1 Introduction

Recently, large language models (LLMs) [BCE⁺23, BMR⁺20, TLI⁺23, JSM⁺23] have achieved remarkable performance on various language-related tasks. However, despite their success in math reasoning [KGR⁺23], common sense reasoning [LKH⁺22], and other reasoning tasks such as symbolic reasoning or logic reasoning [KGR⁺23], their abilities in spatial reasoning still remain underexplored [RFD⁺21, YBL⁺23, MHV⁺24].

Spatial reasoning is an essential function of human cognition, allowing us to interact with the environment. It facilitates tasks that require understanding and reasoning about the spatial relationships between objects and their motions. The spatial reasoning of language models largely relies on language to reason about spatial information, whereas human cognitive capabilities extend far beyond verbal reasoning. Humans can not only create task-relevant abstract representations from visual perception [BK18, KC22], but also imagine unseen scenes through their *mind's eye*. It remains a research topic called mental image [She78] in domains of neuroscience, philosophy of mind, and cognitive science. Building upon this cognitive function, humans facilitate spatial reasoning by mental image manipulation, such as navigation [Tol48], mental rotation [SM71], mental paper folding [SF72], and mental simulation [MK09]. Figure 1 illustrates the human process involved in a navigation task. Humans enhance their spatial awareness and inform their decisions by creating mental images of a route, utilizing various sensory inputs such as navigation instructions or a map image. Subsequently, they simulate route planning through the mind's eye.

Inspired by this cognitive mechanism, we conjecture that LLMs possess the ability to create and manipulate *mental images* in the mind's eye for spatial reasoning. As illustrated in Figure 1, LLMs could potentially process and understand spatial information in various formats. They might be capable of visualizing internal states and manipulating these *mental images* through their *mind's eye*, thereby guiding subsequent reasoning steps to enhance spatial reasoning. Therefore, we propose the **Visualization-of-Thought (VoT)** prompting to elicit this ability. This method leverage LLMs to visualize their reasoning steps and inform subsequent steps, implementing the concept of visuospatial sketchpad [Bad92]. VoT adopts zero-shot prompting instead of relying on few-shot demonstrations or text-to-image visualization with CLIP [RKH⁺21]. This choice stems from LLMs' ability to acquire various *mental images* from text-based visual art [SB14, SMM21, Reg19].

To evaluate the effectiveness of **VoT** in spatial reasoning, we selected three tasks that require spatial awareness in LLMs, including natural-language navigation [YBL⁺23], visual navigation, and visual tiling. These tasks require an understanding of space, direction, and geometric shape reasoning. To emulate human-like multisensory perception, we designed 2D grid worlds using special characters as enriched input formats for the LLMs in visual navigation and visual tiling tasks. We compared different models (GPT-4, GPT-4V) and prompting techniques across these three tasks. The findings reveal that the VoT prompting proposed in this paper consistently induces LLMs to visualize their reasoning steps and inform subsequent steps. Consequently, this approach achieved significant performance improvements on the corresponding tasks.

The main contributions of this paper include:

1. We shed light on LLMs' *mental image* for spatial reasoning from a cognitive perspective, conducting quantitative and qualitative analyses on the mind's eye of LLMs and its limitations. We also explore cues about the origin of this generalized ability from code pre-training.
2. We develop two tasks of "visual navigation" and "visual tiling", along with corresponding synthetic datasets, emulating various sensory inputs for LLMs. These tasks are structured to support varying levels of difficulty, offering a well-designed testbed for the research on spatial reasoning.
3. We propose **Visualization-of-Thought (VoT)** prompting to elicit the mind's eye of LLMs for spatial reasoning and provide empirical evaluations on three tasks. Experiment results prove the effectiveness of VoT prompting compared with other prompting methods and existing MLLMs. This ability to generate *mental images* to facilitate spatial reasoning resembles the mind's eye process, suggesting its potential viability in MLLMs.

2 Spatial Reasoning

Spatial reasoning refers to the ability to comprehend and reason about the spatial relationships among objects, their movements, and interactions with the environment. This skill is vital for a wide range of real-world applications such as navigation, robotics, and autonomous driving. These fields necessitate action planning based on visual perception and a concrete understanding of spatial dimensions.

Although several tasks and datasets [WBC⁺15, SZL22, MK22, LB18, RAB⁺20] have been developed to probe the spatial semantics embedded in text, existing research efforts often focus on how spatial terms are linguistically structured. Recently, significant achievements and impressive results have been achieved in these benchmarks by converting spatial terms to logical forms through LLMs and adopting logic programming [YIL23]. This implies that excelling in these tasks does not necessarily equate to a genuine understanding of spatial information by LLMs, nor does it provide an accurate measure of their spatial awareness.

Spatial awareness involves understanding spatial relationships, directions, distances, and geometric shapes, all of which are essential for action planning in the physical world. To evaluate the spatial awareness and spatial reasoning abilities of LLMs, we have selected tasks that test navigation and geometric reasoning skills, including natural language navigation, visual navigation and visual tiling.

2.1 Natural Language Navigation

Natural language navigation task [YBL⁺23] was inspired by prior research on human cognition [GDB17] presenting participants with sequential transitions sampled from a graph structure.

In this context, a square map is defined by a sequence of random walk instructions and associated objects at each location, denoted as $W = \{(l_1, o_1), (l_2, o_2), \dots, (l_n, o_n)\}$. Given a square map W , and sequence of navigation instructions $I = \{i_1, \dots, i_k\}$, the task for the model is to identify the associated object $o \in W$ at the specified location l which is determined by the navigation instructions, as detailed in Equation 1 and exemplified in Appendix B.2.

$$o \sim p(o \in W | W = \{(l_1, o_1), (l_2, o_2), \dots, (l_n, o_n)\}, I) \quad (1)$$

2.2 Visual Navigation

Visual navigation task presents a synthetic 2D grid world to LLM, challenging it to navigate using visual cues. The model must generate navigation instructions to move in four directions (left, right, up, down) to reach the destination from the starting point while avoiding obstacles. This involves two sub-tasks: **route planning** and **next step prediction**, requiring multi-hop spatial reasoning, while the former is more complex. Task instructions are available in Figure 6 in appendix.

Formulation The model is presented with a grid map M consisting of k consecutive edges $E = \{e(s_0, s_1), e(s_1, s_2), \dots, e(s_{k-1}, s_k)\}$, where the starting point and destination are s_0 and s_k respectively, as shown in Figure 2. Route planning task is to generate a sequence of correct directions $D = \{d(s_0, s_1), d(s_1, s_2), \dots, d(s_{k-1}, s_k)\}$, as defined in Equation 2. Given M and t navigation instructions $D_{t, 0 < t < k} = \{d(s_0, s_1), \dots, d(s_{t-1}, s_t)\}$, next step prediction task is to identify the correct direction $d(s_t, s_{t+1})$ of the next step, as defined in Equation 3.

$$D \sim p(\{d(s_0, s_1), d(s_1, s_2), \dots, d(s_{k-1}, s_k)\} | M) \quad (2)$$

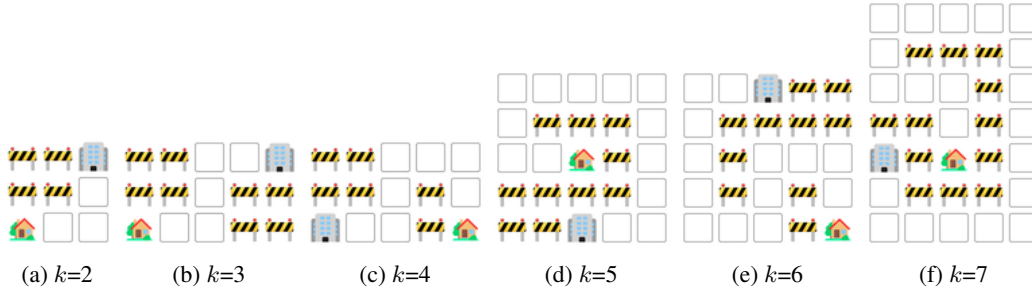


Figure 2: Examples of a navigation map under different settings of k , with emoji of house indicating the starting point, and emoji of office indicating the destination.